

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 7. String and its Methods

Lecture: 73. find() and index() Methods

Section 1: Lecture Summary

This lecture focused on several important Python string methods used to find the position of substrings and count occurrences within strings. The instructor introduced five main methods of the built-in `str` class: `find()`, `rfind()`, `index()`, `rindex()`, and `count()`. Each method searches for a substring in a string but differs slightly in behavior, especially regarding the direction of search and how they handle not-found cases. The lecture demonstrated how to call these methods, their parameters (substring, optional start index, and optional end index), and illustrated their behavior with examples using the string `"Hello How are you"`. Key differences between similar methods (`find` vs `index`, `rfind` vs `rindex`) were highlighted, particularly how `find` returns `-1` if not found while `index` raises an error. The lecture concluded with encouragement to practice these methods with different strings for mastery.

Section 2: Key Concepts and Explanations

- String Methods Overview

- Strings in Python are objects of the `str` class.
- The `str` class includes various methods to manipulate and search strings.
- This lecture covers methods related to finding substring indices and counting substring occurrences.

- Methods for Finding Substrings

- `find(sub, start=0, end=len(string))`
 - Searches for the substring `sub` within the string.
 - Returns the lowest index where `sub` is found.

- Optional `start` and `end` specify the slice of the string to search in.

- Returns `-1` if `sub` is not found.

- Example: `s.find('o')` returns `4` for `"Hello How are you"`.

- **`rfind(sub, start=0, end=len(string))`**

- Similar to `find` but searches from the right (end) of the string.

- Returns the highest index where `sub` is found.

- Returns `-1` if not found.

- Example: `s.rfind('o')` returns `15`.

- **`index(sub, start=0, end=len(string))`**

- Functions like `find` but raises a `ValueError` exception if `sub` is not found instead of returning `-1`.

- Useful when missing substring should be handled explicitly.

- Example: `s.index('o')` returns `4`, but `s.index('k')` raises an error.

- **`rindex(sub, start=0, end=len(string))`**

- Same as `rfind` but raises an exception if not found.

- Searches from the right end of the string.

- Example: `s.rindex('o')` returns `15`.

- Counting Occurrences

- **`count(sub, start=0, end=len(string))`**

- Counts how many times `sub` occurs in the string or its substring slice.

- Returns 0 if not found.

- Example: `s.count('o')` returns `3`.

- Parameters Common to These Methods

- `sub` (mandatory): The substring to search or count within the string.

- ``start`` (optional): The starting index to begin the search/count, inclusive.

- ``end`` (optional): The ending index to stop the search/count, exclusive.

- Behavior When Substring Not Found

- ``find``, ``rfind``, and ``count`` methods return ``-1`` or ``0`` gracefully.

- ``index`` and ``rindex`` methods raise exceptions (``ValueError``), requiring error handling.

Section 3: Example Code and Use Cases

Using `find()`

Using `find()` with start parameter

Using `find()` with start and end parameters

Using `rfind()`

Using `index()`

The following will raise `ValueError`:

```
x = s.index('k')
```

Using `rindex()`

Using `count()`

```
s = "Hello How are you"

x = s.find('o') # Finds first occurrence of 'o'
print(x) # Output: 4

x = s.find('how') # Case-sensitive search
print(x) # Output: 6

x = s.find('k') # Substring not present
print(x) # Output: -1

x = s.find('o', 5) # Search for 'o' starting from index 5
print(x) # Output: 7
```

```
x = s.find('o', 5, 7) # Search in slice s[5:7]
print(x) # Output: -1

x = s.rfind('o') # Finds last occurrence of 'o'
print(x) # Output: 15

x = s.rfind('o', 0, 15) # Search from the right up to index 15
print(x) # Output: 7

x = s.index('o') # Like find(), but raises error if not found
print(x) # Output: 4

x = s.index('how') # Raises error if substring not found
print(x) # Output: 6

x = s.rindex('o') # Like rfind(), but raises error if not found
print(x) # Output: 15

x = s.count('o') # Counts number of 'o' occurrences
print(x) # Output: 3

x = s.count('me') # Substring not present
print(x) # Output: 0
```

Explanation:

- This code demonstrates calls to each method, shows handling of optional parameters, and outputs returned indices or counts.
- It illustrates difference in error behavior between `find` and `index` methods.
- Using slicing parameters allows flexible substring searches within defined parts of the larger string.

Section 4: Key Takeaways

- Python string methods `find`, `rfind`, `index`, `rindex`, and `count` help locate and count substrings.
- `find` and `rfind` return `-1` if substring is not found; `index` and `rindex` raise exceptions.
- Optional start and end indices let you restrict search or count operations to string slices.

- ``rfind`` and ``rindex`` search from right to left, useful for last occurrence queries.
- Knowing these methods and their differences is critical for effective string manipulation in Python.
- Practice using these methods with various strings and substrings to memorize their usage and behavior.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.





Find & Index



Find & Index
String Methods

Find & Index

String Methods

find(sub, start, end)

Find & Index

String Methods

find(sub, start, end)

S

H	e	l	l	o		H	o	w		a	r	e		Y	o	u
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

S

H	e	l	l	o		H	o	w		a	r	e		Y	o	u
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

S H e l l o H o w a r e Y o u
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

S H e l l o H o w a r e Y o u
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

S

H	e	l	l	o		H	o	w		a	r	e		Y	o	u
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

Find & Index

String Methods

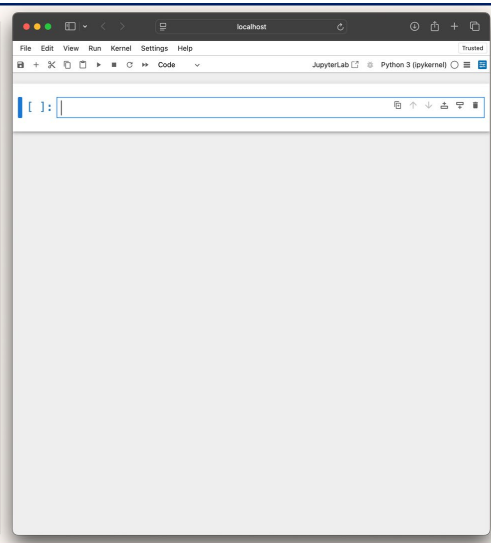
`find( sub, start, end )`

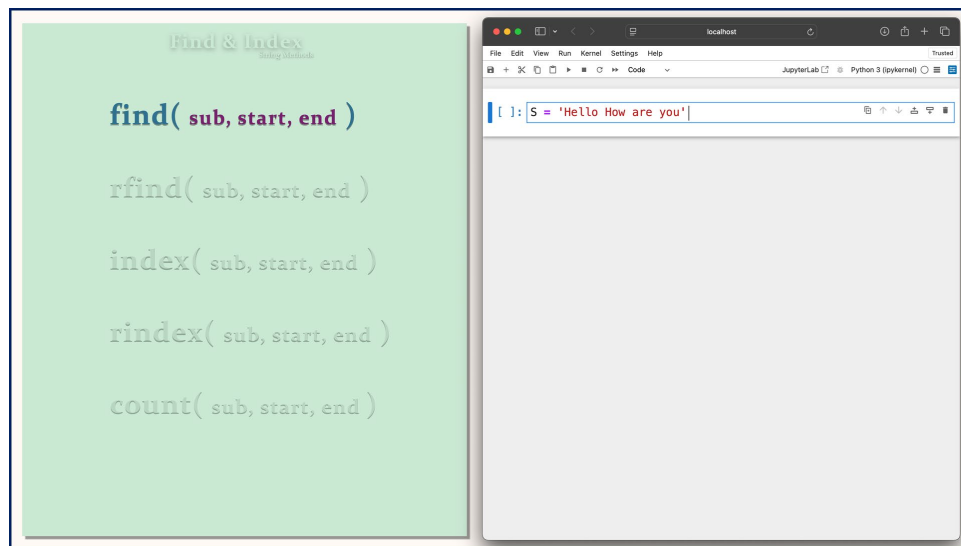
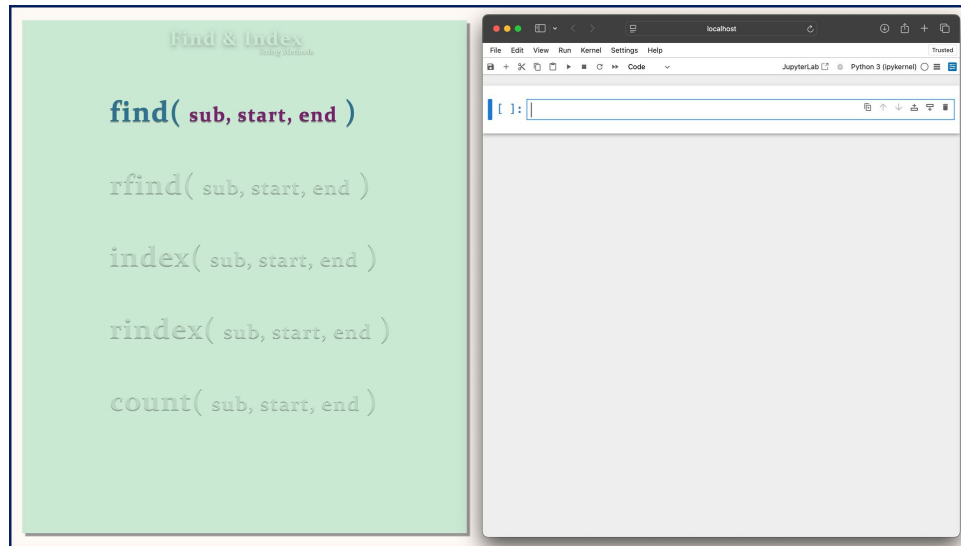
`rfind( sub, start, end )`

`index( sub, start, end )`

`rindex( sub, start, end )`

`count( sub, start, end )`





Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

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JupyterLab Python 3 (ipykernel)

[1]: S = 'Hello How are you'

[]:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

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Trusted

JupyterLab Python 3 (ipykernel)

[1]: S = 'Hello How are you'

[]:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

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JupyterLab Python 3 (ipykernel)

```
[1]: S = 'Hello How are you'
```

```
[ ]:
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

S

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

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JupyterLab Python 3 (ipykernel)

```
[1]: S = 'Hello How are you'
```

```
[ ]:
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

S.find('o')

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

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Trusted

Python 3 (ipykernel)

[1]: S = 'Hello How are you'

[]:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

x = S.find('o')

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

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Trusted

Python 3 (ipykernel)

[1]: S = 'Hello How are you'

[]: X = S.find('o')

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

x = S.find('o')

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

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Trusted

JupyterLab Python 3 (ipykernel)

[1]: S = 'Hello How are you'

[2]: X = S.find('o')

[]:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

X 4

x = S.find('o')

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

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Trusted

JupyterLab Python 3 (ipykernel)

[1]: S = 'Hello How are you'

[2]: X = S.find('o')

[3]: print(X)

4

[]:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

X 4

x = S.find('o')

Find & Index

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

The screenshot shows a JupyterLab window with a Python 3 (ipykernel) environment. The code cell contains the following code:

```
[1]: S = 'Hello How are you'
```

The output cell displays the string 'Hello How are you' with its index mapping. The string is shown as a sequence of characters in a table, with indices 0 through 16. The characters 'H', 'o', 'w' are highlighted in red, and a bracket is drawn under them, indicating the substring 'How'.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

```
[1]: S = 'Hello How are you'
```

```
[ ]:
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
S	H	e	l	l	o	H	o	w	a	r	e	Y	o	u		

S.find()

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

```
[1]: S = 'Hello How are you'
```

```
[ ]:
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
S	H	e	l	l	o	H	o	w	a	r	e	Y	o	u		

S.find('How')

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

[1]: S = 'Hello How are you'

[]:

S H e l l o H o w a r e Y o u

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

x = S.find('How')

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

[1]: S = 'Hello How are you'

[]: x = S.find('How')

S H e l l o H o w a r e Y o u

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

x = S.find('How')

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

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JupyterLab Python 3 (ipykernel)

```
[1]: S = 'Hello How are you'
[4]: X = S.find('How')
```

```
[ ]:
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
S	H	e	l	l	o	H	o	w	a	r	e	Y	o	u		

X 6

x = S.find('How')

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

File Edit View Run Kernel Settings Help

JupyterLab Python 3 (ipykernel)

```
[1]: S = 'Hello How are you'
[4]: X = S.find('How')
[5]: print(X)
```

```
6
[ ]:
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
S	H	e	l	l	o	H	o	w	a	r	e	Y	o	u		

X 6

x = S.find('How')

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

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Trusted

Python 3 (ipykernel)

[1]: S = 'Hello How are you'

[]:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

x = S.find('k')

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

File Edit View Run Kernel Settings Help

Trusted

Python 3 (ipykernel)

[1]: S = 'Hello How are you'

[]: X = S.find('k')

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

x = S.find('k')

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python 3 (ipykernel)

```
[1]: S = 'Hello How are you'
[6]: X = S.find('k')
```

[]:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

X -1

x = S.find('k')

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python 3 (ipykernel)

```
[1]: S = 'Hello How are you'
[6]: X = S.find('k')
[7]: print(X)
```

[]:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

X -1

x = S.find('k')

Find & Index

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

The screenshot shows a JupyterLab window with a code editor and a console. The code editor contains the following Python code:

```
[1]: S = 'Hello How are you'
```

The console output shows the string 'Hello How are you' with its index representation below it:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

```
[1]: S = 'Hello How are you'
```

```
[ ]:
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

`x = S.find('o')`

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

```
[1]: S = 'Hello How are you'
```

```
[ ]:
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

`x = S.find('o',5)`

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python 3 (ipykernel)

[1]: S = 'Hello How are you'

[]: X = S.find('o',5)

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

x = S.find('o',5)

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python 3 (ipykernel)

[1]: S = 'Hello How are you'

[6]: X = S.find('o',5)

[]:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

X 7

x = S.find('o',5)

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python 3 (ipykernel)

```
[1]: S = 'Hello How are you'
[8]: X = S.find('o',5)
[9]: print(X)
7
```

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

S H e l l o H o w a r e Y o u

X 7

x = S.find('o',5)

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python 3 (ipykernel)

```
[1]: S = 'Hello How are you'
[ ]:
```

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

S H e l l o H o w a r e Y o u

o

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

```
[1]: S = 'Hello How are you'
```

```
[ ]:
```

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
S	H	e	l	l	o	H	o	w	a	r	e	Y	o	u			

x = S.find('o',5,7)

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

```
[1]: S = 'Hello How are you'
```

```
[ ]: X = S.find('o', 5, 7)
```

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
S	H	e	l	l	o	H	o	w	a	r	e	Y	o	u			

x = S.find('o',5,7)

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python 3 (ipykernel)

[1]: S = 'Hello How are you'

[10]: X = S.find('o', 5, 7)

[]:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
S	H	e	l	l	o	H	o	w	a	r	e	Y	o	u		

X -1

o

x = S.find('o',5,7)

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python 3 (ipykernel)

[1]: S = 'Hello How are you'

[10]: X = S.find('o', 5, 7)

[11]: print(X)

-1

[]:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
S	H	e	l	l	o	H	o	w	a	r	e	Y	o	u		

X -1

o

x = S.find('o',5,7)

Find & Index

String Methods

`find( sub, start, end )`

`rfind( sub, start, end )`

`index( sub, start, end )`

`rindex( sub, start, end )`

`count( sub, start, end )`

The screenshot shows a JupyterLab window with a Python 3 (ipykernel) environment. The code cell contains the assignment `S = 'Hello How are you'`. Below the code, the string is displayed with its characters and their corresponding indices from 0 to 16. The characters are: S, H, e, l, l, o, H, o, w, a, r, e, Y, o, u. The indices are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16. The characters are color-coded: 'S' is black, 'H' is red, 'e' is blue, 'l' is green, 'l' is green, 'o' is blue, 'H' is red, 'o' is blue, 'w' is green, 'a' is blue, 'r' is green, 'e' is blue, 'Y' is red, 'o' is blue, and 'u' is green.

```
[1]: S = 'Hello How are you'
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
S	H	e	l	l	o	H	o	w	a	r	e	Y	o	u		

Find & Index

String Methods

`find( sub, start, end )`

`rfind( sub, start, end )`

`index( sub, start, end )`

`rindex( sub, start, end )`

`count( sub, start, end )`

```
[1]: S = 'Hello How are you'
```

```
[ ]:
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

Find & Index

String Methods

`find( sub, start, end )`

`rfind( sub, start, end )`

`index( sub, start, end )`

`rindex( sub, start, end )`

`count( sub, start, end )`

```
[1]: S = 'Hello How are you'
```

```
[ ]: X = S.rfind('o')
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

`x = S.rfind('o')`

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

File Edit View Run Kernel Settings Help

Python 3 (ipykernel)

[1]: S = 'Hello How are you'

[12]: X = S.rfind('o')

[]:

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

S H e l l o H o w a r e Y o u

X

15

X = S.rfind('o')

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

File Edit View Run Kernel Settings Help

Python 3 (ipykernel)

[1]: S = 'Hello How are you'

[12]: X = S.rfind('o')

[13]: print(X)

15

[]:

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

S H e l l o H o w a r e Y o u

X

15

X = S.rfind('o')

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

File Edit View Run Kernel Settings Help

JupyterLab Python 3 (ipykernel)

[1]: S = 'Hello How are you'

[]: X = S.rfind('o', 0, 15)

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

x = S.rfind('o',0,15)

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

File Edit View Run Kernel Settings Help

JupyterLab Python 3 (ipykernel)

[1]: S = 'Hello How are you'

[14]: X = S.rfind('o', 0, 15)

[]:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

x 7

x = S.rfind('o',0,15)

Find & Index

String Methods

```
find( sub, start, end )
```

```
rfind( sub, start, end )
```

```
index( sub, start, end )
```

```
rindex( sub, start, end )
```

```
count( sub, start, end )
```

```
[1]: S = 'Hello How are you'
```

```
[14]: X = S.rfind('o', 0, 15)
```

```
[15]: print(X)
```

```
7
```

```
[ ]:
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

X 7

```
X = S.rfind('o',0,15)
```

Find & Index

String Methods

```
find( sub, start, end )
```

```
rfind( sub, start, end )
```

```
index( sub, start, end )
```

```
rindex( sub, start, end )
```

```
count( sub, start, end )
```

```
[1]: S = 'Hello How are you'
```

```
[ ]: X = S.rfind('kite')
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

```
X = S.rfind('kite')
```

Find & Index

String Methods

`find( sub, start, end )`

`rfind( sub, start, end )`

`index( sub, start, end )`

`rindex( sub, start, end )`

`count( sub, start, end )`

localhost

File Edit View Run Kernel Settings Help

Trusted

Python 3 (ipykernel)

```
[1]: S = 'Hello How are you'
[16]: X = S.rfind('kite')
[ ]:
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

X

-1

X = S.rfind('kite')

Find & Index

String Methods

`find( sub, start, end )`

`rfind( sub, start, end )`

`index( sub, start, end )`

`rindex( sub, start, end )`

`count( sub, start, end )`

localhost

File Edit View Run Kernel Settings Help

Trusted

Python 3 (ipykernel)

```
[1]: S = 'Hello How are you'
[16]: X = S.rfind('kite')
[17]: print(X)
[ ]:
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

X

-1

X = S.rfind('kite')

Find & Index

String Methods

`find( sub, start, end )`

`rfind( sub, start, end )`

`index( sub, start, end )`

`rindex( sub, start, end )`

`count( sub, start, end )`

The screenshot shows a JupyterLab window with a Python 3 (ipykernel) environment. The code cell contains the assignment `S = 'Hello How are you'`. Below the code, the string is displayed with its characters and their corresponding indices from 0 to 16. The characters are: H (0), e (1), l (2), l (3), o (4), H (5), o (6), w (7), a (8), r (9), e (10), Y (11), o (12), u (13). The string ends with a space character at index 14, which is not visible in the provided image.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
S	H	e	l	l	o	H	o	w	a	r	e	Y	o	u		

Find & Index

String Methods

```
find( sub, start, end )  
rfind( sub, start, end )  
index( sub, start, end )  
rindex( sub, start, end )  
count( sub, start, end )
```

```
[1]: S = 'Hello How are you'  
[ ]:
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

Find & Index

String Methods

```
find( sub, start, end )  
rfind( sub, start, end )  
index( sub, start, end )  
rindex( sub, start, end )  
count( sub, start, end )
```

```
[1]: S = 'Hello How are you'  
[ ]: X = S.index('o')
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

`x = S.index('o')`

Find & Index

String Methods

`find( sub, start, end )`

`rfind( sub, start, end )`

`index( sub, start, end )`

`rindex( sub, start, end )`

`count( sub, start, end )`

localhost

File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python 3 (ipykernel)

```
[1]: S = 'Hello How are you'
[18]: X = S.index('o')
[ ]:
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

X **4**

`x = S.index('o')`

Find & Index

String Methods

`find( sub, start, end )`

`rfind( sub, start, end )`

`index( sub, start, end )`

`rindex( sub, start, end )`

`count( sub, start, end )`

localhost

File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python 3 (ipykernel)

```
[1]: S = 'Hello How are you'
[18]: X = S.index('o')
[19]: print(X)
4
[ ]:
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

X **4**

`x = S.index('o')`

Find & Index

String Methods

```
find( sub, start, end )
```

```
rfind( sub, start, end )
```

```
index( sub, start, end )
```

```
rindex( sub, start, end )
```

```
count( sub, start, end )
```

localhost

File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python 3 (ipykernel)

[1]: S = 'Hello How are you'

[]:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

```
x = S.index('How')
```

Find & Index

String Methods

```
find( sub, start, end )
```

```
rfind( sub, start, end )
```

```
index( sub, start, end )
```

```
rindex( sub, start, end )
```

```
count( sub, start, end )
```

localhost

File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python 3 (ipykernel)

[1]: S = 'Hello How are you'

[]: X = S.index('How')

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

```
x = S.index('How')
```

Find & Index

String Methods

```
find( sub, start, end )  
rfind( sub, start, end )  
index( sub, start, end )  
rindex( sub, start, end )  
count( sub, start, end )
```

```
[1]: S = 'Hello How are you'  
[20]: X = S.index('How')  
[ ]:
```

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
S H e l l o H o w a r e Y o u
X 6
X = S.index('How')

Find & Index

String Methods

```
find( sub, start, end )  
rfind( sub, start, end )  
index( sub, start, end )  
rindex( sub, start, end )  
count( sub, start, end )
```

```
[1]: S = 'Hello How are you'  
[20]: X = S.index('How')  
[21]: print(X)  
[ ]: 6  
[ ]:
```

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
S H e l l o H o w a r e Y o u
X 6
X = S.index('How')

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python 3 (ipykernel)

[1]: S = 'Hello How are you'

[]: X = S.index('k')

S H e l l o H o w a r e Y o u

x = S.index('k')

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python 3 (ipykernel)

[1]: S = 'Hello How are you'

[22]: X = S.index('k')

ValueError: substring not found

Traceback (most recent call last):

Cell In[22], line 1

1 X = S.index('k')

[]:

S H e l l o H o w a r e Y o u

x = S.index('k')

Find & Index

String Methods

`find( sub, start, end )`

`rfind( sub, start, end )`

`index( sub, start, end )`

`rindex( sub, start, end )`

`count( sub, start, end )`

The screenshot shows a JupyterLab window with a Python 3 (ipykernel) environment. The code cell contains the assignment `S = 'Hello How are you'`. Below the code, the string is displayed with its characters and indices. The string is "Hello How are you" and the indices are 0 through 16. The characters are: H (0), e (1), l (2), l (3), o (4), H (5), o (6), w (7), a (8), r (9), e (10), Y (11), o (12), u (13). The string is displayed in a monospace font with a light blue background.

```
[1]: S = 'Hello How are you'
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
S	H	e	l	l	o	H	o	w	a	r	e	Y	o	u		

Find & Index

String Methods

```
find( sub, start, end )  
rfind( sub, start, end )  
index( sub, start, end )  
rindex( sub, start, end )  
count( sub, start, end )
```

```
[1]: S = 'Hello How are you'  
[ ]: X = S.rindex('o')
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

X = S.rindex('o')

Find & Index

String Methods

```
find( sub, start, end )  
rfind( sub, start, end )  
index( sub, start, end )  
rindex( sub, start, end )  
count( sub, start, end )
```

```
[1]: S = 'Hello How are you'  
[23]: X = S.rindex('o')  
[ ]:
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

X 15

X = S.rindex('o')

Find & Index

String Methods

```
find( sub, start, end )  
rfind( sub, start, end )  
index( sub, start, end )  
rindex( sub, start, end )  
count( sub, start, end )
```

```
[1]: S = 'Hello How are you'  
[23]: X = S.rindex('o')  
[24]: print(X)  
15
```

S H e l l o H o w a r e Y o u

X 15

X = S.rindex('o')

Find & Index

String Methods

```
find( sub, start, end )  
rfind( sub, start, end )  
index( sub, start, end )  
rindex( sub, start, end )  
count( sub, start, end )
```

```
[1]: S = 'Hello How are you'  
[ ]: X = S.rindex('o', 0, 15)
```

S H e l l o H o w a r e Y o u

X = S.rindex('o',0,15)

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

File Edit View Run Kernel Settings Help

JupyterLab Python 3 (pykernel)

[1]: S = 'Hello How are you'

[25]: X = S.rindex('o', 0, 15)

[]:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

X 7

x = S.rindex('o',0,15)

Find & Index

String Methods

find(sub, start, end)

rfind(sub, start, end)

index(sub, start, end)

rindex(sub, start, end)

count(sub, start, end)

localhost

File Edit View Run Kernel Settings Help

JupyterLab Python 3 (pykernel)

[1]: S = 'Hello How are you'

[25]: X = S.rindex('o', 0, 15)

[26]: print(X)

7

[]:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

X 7

x = S.rindex('o',0,15)

Find & Index

String Methods

```
find( sub, start, end )  
rfind( sub, start, end )  
index( sub, start, end )  
rindex( sub, start, end )  
count( sub, start, end )
```

localhost

File Edit View Run Kernel Settings Help

JupyterLab Python 3 (ipykernel)

```
[1]: S = 'Hello How are you'  
[ ]:
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
S	H	e	l	l	o	H	o	w	a	r	e	Y	o	u		

Find & Index

String Methods

```
find( sub, start, end )  
rfind( sub, start, end )  
index( sub, start, end )  
rindex( sub, start, end )  
count( sub, start, end )
```

localhost

File Edit View Run Kernel Settings Help

JupyterLab Python 3 (ipykernel)

```
[1]: S = 'Hello How are you'  
[ ]: X = S.count('o')  
[ ]:
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
S	H	e	l	l	o	H	o	w	a	r	e	Y	o	u		

X = S.count('o')

Find & Index

String Methods

```
find( sub, start, end )  
rfind( sub, start, end )  
index( sub, start, end )  
rindex( sub, start, end )  
count( sub, start, end )
```

```
[1]: S = 'Hello How are you'  
[2]: X = S.count('o')
```

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

X 3

X = S.count('o')

Find & Index

String Methods

```
find( sub, start, end )  
rfind( sub, start, end )  
index( sub, start, end )  
rindex( sub, start, end )  
count( sub, start, end )
```

```
[1]: S = 'Hello How are you'  
[2]: X = S.count('o')  
[3]: print(X)  
3
```

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

X 3

X = S.count('o')

Find & Index

String Methods

`find( sub, start, end )`

`rfind( sub, start, end )`

`index( sub, start, end )`

`rindex( sub, start, end )`

`count( sub, start, end )`

```
[1]: S = 'Hello How are you'
[4]: X = S.count('me')
```

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

S H e l l o H o w a r e Y o u

X 0

`X = S.count('me')`

Find & Index

String Methods

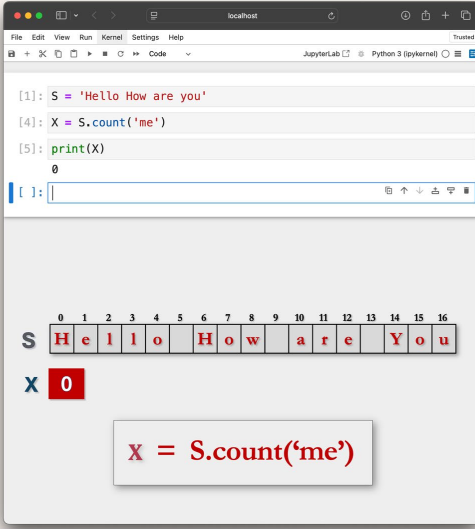
`find( sub, start, end )`

`rfind( sub, start, end )`

`index( sub, start, end )`

`rindex( sub, start, end )`

`count( sub, start, end )`



The screenshot shows a JupyterLab window with the following content:

```
[1]: S = 'Hello How are you'
[4]: X = S.count('me')
[5]: print(X)
0
```

Below the code, the output is displayed as a string 'S' followed by a table of characters and their indices:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
S	H	e	l	l	o		H	o	w		a	r	e		Y	o	u

Below the table, the variable `X` is shown with its value `0`:

`X 0`

At the bottom, a box contains the code `X = S.count('me')`.